# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours) Regular Examination, October 2025 Third Semester

**Microbiology and Phycology (DSC) -MG3DSCBOT200**

**Part A**
**Short answer type questions.**

**Answer any 10 questions. Each question carries 1 mark**

1. What are pili?

   They are thin, hair-like appendages seen on the surface of many bacteria.

2. What is the function of bacterial flagella?

   These are the organs of locomotion

3. What is the function of the tail fibre in λ phage?

   It helps to specifically bind to the receptor sites of a bacterial surface

4. Mention one microorganism that plays a significant role in environmental conservation.

   Name any one nitrogen-fixing bacteria or biodegrading bacteria eg: *Pseudomonas putida*,

5. Define coenobium

   A coenobium is a loose association of unicellular organisms that lives in a colony

6. State two major features of Lee's 2016 system of classification in algae.

   - Use of phylogenetic relationships among algae, based on molecular data to group algae
   - Based on the ultrastructural characteristics of chloroplasts within different algal groups

7. Which structure in *Sargassum* helps it float?

   Air-filled bladders

8. Name the haploid generation in the life cycle of algae?

   Gametophyte

9. Which economically valuable product is extracted from *Gelidium* and *Gracilaria*?

   Agar-agar or any other relevant product

10. What is SCP?

Single Cell Protein- It is the edible protein extracted from unicellular microorganisms like bacteria, fungi or algae

11. What is a photobioreactor in algal cultivation?

It is a closed, controlled system that uses artificial or solar light to grow algae

12. Name two commercial products obtained from macro algae

Agar, carrageenin, algins, diatomite etc. (Any two)

**Part B**

**Short Essay type questions.**

**Answer any 6 questions. Each question carries 5 marks**

13. Discuss the advantages of biopesticides over chemical pesticides

-Environmental safety, target specific, reduced toxicity, pests are less resistant and improved soil health. **(Liberal valuation)**

14. Distinguish between generalized and specialized transduction

The process of genetic recombination occurs when a bacteriophage (phage virus) transfers DNA from one bacterium to another.

Generalized Transduction

- Occurs during the lytic cycle of a phage.
- Any random gene from the donor bacterium may be transferred.
- Happens when the phage mistakenly packages bacterial DNA into its capsid.
- Occurs frequently in the phage life cycle.

Specialized Transduction

- Occurs during the lysogenic cycle of a temperate phage.
- Only specific genes near the phage integration site are transferred.
- Happens when the phage incorrectly excises from the bacterial chromosome, taking adjacent genes with it.
- Occurs rarely compared to generalized transduction.

15. Write short notes on terrestrial and aquatic forms of algae

**Aquatic Algae**: Found in water bodies such as ponds, lakes, rivers, and seas. May be free-floating (planktonic), swimming (motile), or attached to rocks or submerged surfaces. Include both freshwater and marine forms.(Give any one example for each).

**Terrestrial Algae**: Occur on land, particularly in moist or damp habitats such as soil, tree trunks, rocks, and walls. Based on habitat, terrestrial algae are divided into: a. Saprophytic Forms: Grow on damp soil, rocks, or tree surfaces.eg; *Vaucheria* b. Cryptophytic Forms: Grow

inside the soil and contribute to nitrogen fixation and soil fertility. Examples: *Nostoc*, *Anabaena, Trentipohlia* etc*)* Symbiotic Forms: Some algae form symbiotic associations with fungi to produce lichens, which can grow on rocks, tree bark, and walls.
(**Examples can also be considered for marks)**

16. Describe the main features of the vegetative structure in *Chara.*

- The thallus is macroscopic, erect, and plant-like.
- It is clearly differentiated into rhizoids, main axis, and branches.
- The main axis has nodes and internodes
- Branches of unlimited growth and limited growth
- Cell wall made of cellulose, encrusted with calcium carbonate, giving a stony texture.
- Discoid Chloroplast with pyrenoids storing starch

17. Describe the main features of a *Cladophora* cell

- Each cell is large and multinucleated (coenocytic).
- The cell wall is thick, composed of outer chitin, middle pectic, and inner cellulosic layers.
- Reticulate chloroplast with several pyrenoids
- The cytoplasm forms a thin lining along the inner wall and encloses a large central vacuole.

18. Give a brief overview of the evolutionary sequence of algae, as suggested by Fritsch

- The progression generally moves from simple to more complex forms
- Unicellular motile forms - Unicellular non-motile (coccoid) forms- Colonial forms- Filamentous forms- Heterotrichous forms- Parenchymatous forms
- The eleven classes defined by Fritsch reflect these evolutionary grades. (e.g., write habit of Chlorophyceae, Phaeophyceae, and Rhodophyceae)- **(Liberal valuation)**

19. Discuss the significance of algal-fungal symbiosis in lichens

Lichens are the best example of a symbiotic association between algae and fungi. Fungus provides structure and protection, while alga performs photosynthesis.
It has the following ecological roles
Soil formation, Bioindicators, and Nutrient cycling
**( Only very brief description needed)**

20. How do algae contribute to global primary production?

- Photosynthesis – primary producers- life on earth through food web
- Contributes to half of the world's production and 70% of the earth's oxygen
- CO2 absorption

## Part C

### Essay type questions.
### Answer any 1 question, question carries 5 marks

21. Compare and contrast viruses, viroids and prions

Viruses, viroids and prions are acellular infectious agents that cannot reproduce on their own. They depend entirely on living host cells for multiplication.

Viruses

- Non-cellular particles made of nucleic acid (DNA or RNA) enclosed in a protein coat (capsid). Some have an envelope and spikes.
- Ultramicroscopic, about 20–300 nm.
- No cytoplasm or organelles.
- Obligate intracellular parasites
- Hosts: Infect plants, animals, and bacteria (bacteriophages). Examples: *Tobacco Mosaic Virus (TMV)*,

2. Viroids

- Smallest infectious agents made of short, circular, single-stranded RNA (250–400 nucleotides).
- Infect only plants.
- Multiply in the nucleus or chloroplast using host enzymes.
- Interfere with gene expression, causing diseases like stunted growth and deformation. Example: *Potato Spindle Tuber Viroid (PSTV)*.

3. Prions

- Infectious protein particles without DNA or RNA.
- Abnormal forms of normal nerve-cell proteins.
- Convert normal proteins into abnormal forms, leading to the degeneration of brain tissue.
- Extremely resistant to heat, radiation, and disinfectants.
- Diseases: Cause neurological disorders in animals and humans (e.g., *Mad cow disease*, *Creutzfeldt–Jakob disease*).

**(General characteristics with examples for each group can be considered for marks)**

22. What are the advantages and disadvantages of water blooms?

**(Liberal valuation. Anything related to algal blooms can be considered.**

| Advantages | Disadvantages |
|---|---|
| Nitrogen-fixing algae increase fertility | Oxygen depletion due to decomposition of dense blooms |
| Serve as food for birds and aquatic animals. | Produce toxins (e.g., microcystins, anatoxins) harmful to life. |
| Indicate nutrient pollution — useful for environmental monitoring. | Cause bad odour and taste in water. |
| | Clog water filters, irrigation channels, and fish gills.<br>Block sunlight, reducing photosynthesis in submerged plants. |